Customer and Profitability Analysis in Retail

1. Business Problem Statement

A retail company wants to better understand customer purchasing behavior, product performance, profitability trends, and the impact of discounts on business revenue. The company aims to identify profitable customer segments, high-performing products, loss-making categories, and seasonal sales trends to improve decision-making and increase overall profitability.

Main Business Question

How can the company use customer and sales data to improve profitability, identify business risks, and optimize product and discount strategies?

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2. Project Deliverables

1. Data Cleaning & Preparation (Python)

- Cleaned missing values
- Checked duplicates
- Converted date columns
- Standardized column names
- Performed Exploratory Data Analysis (EDA)

2. Business Analysis (SQL)

- Analyzed customer segments
- Identified profitable and loss-making products
- Evaluated discount impact on profitability
- Studied sales trends across months and regions

3. Dashboard Creation (Power BI)

- Built interactive dashboards
- Created KPI cards and trend charts
- Visualized customer and profitability insights

4. Project Documentation

- Summarized business insights
- Added SQL queries and findings
- Documented recommendations for business improvement

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3. Project Overview

This project analyzes retail sales data to understand customer behavior, product profitability, discount impact, regional performance, and sales trends. The goal is to uncover insights that help businesses improve revenue, reduce losses, and make better strategic decisions.

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4. Dataset Summary

Dataset Information

- Dataset Name: Sample - Superstore Dataset
- Rows: 9,994
- Columns: 21

Main Features

- Customer Segment
- Region and State
- Product Category and Sub-Category
- Sales and Profit
- Discount
- Quantity
- Shipping Mode
- Order Date

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5. Exploratory Data Analysis (EDA) Using Python

Data Loading

Imported the dataset using pandas.

```
import pandas as pd
```

Initial Exploration

Used:

- df.info()
- df.describe()
- df.isnull().sum()

To understand:

- dataset structure
- datatypes
- missing values
- summary statistics

Data Cleaning

Missing Value Handling:

- Checked null values
- Handled missing data using appropriate methods

Duplicate Check:

- Removed duplicate rows if present

Date Conversion:

Converted Order Date into datetime format.

```
df['order_date']=pd.to_datetime(df['order_date'])
df['ship_date']=pd.to_datetime(df['ship_date'])
```

Column Standardization:

Renamed columns into snake_case format.

```
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ','_').str.replace('-','_')
```

Feature Engineering

Created New Features

- Year column
- Shipping Days column
- Shipping Status column

```
df['year']=df['order_date'].dt.year
df['shipping_days']=(df['ship_date']-df['order_date']).dt.days
```

```python
def shipping_category(shipping_days):
  if shipping_days<=2:
    return 'Fast delivery'
  elif shipping_days<=6:
    return 'Standard delivery'
  else:
    return 'Delayed delivery'

df['ship_status']=df['shipping_days'].apply(shipping_category)
```

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6. SQL Business Analysis

1. Which customer segments contribute the highest sales and revenue?

SQL Query

```sql
SELECT segment,
    ROUND(SUM(sales),2) AS revenue
FROM retail
GROUP BY segment
ORDER BY revenue DESC;
```

Insight:

The Consumer segment contributes the highest revenue, generating more sales than the Corporate and Home Office segments. This shows that individual customers are the main source of business revenue

Business Meaning:

- Individual customers purchase products more frequently.
- Business depends heavily on Consumer customers.

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2. Which customer segments generate the highest profit?

SQL Query

```sql
SELECT segment,
    ROUND(SUM(profit),2) AS profits
FROM retail
GROUP BY segment
ORDER BY profits DESC;
```

Insight:

The Consumer segment generates the highest profit compared to other customer segments.

Business Meaning:

Consumer customers not only buy more products but also help the company earn higher profits.

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3. Which products have high sales but low profit?

SQL Query

```
SELECT sub_category,
    ROUND(SUM(sales),2) AS sales,
    ROUND(SUM(profit),2) AS profits
FROM retail
GROUP BY sub_category
ORDER BY sales DESC, profits ASC;
```

Insight:

Tables, Machines, Bookcases, and Supplies generate good sales but bring very little profit or even losses for the business.

Business Meaning:

Possible reasons include high discounts, shipping costs, or low pricing strategies.

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4. What are the monthly and yearly sales trends?

SQL Query

```
SELECT YEAR(order_date) AS year,
    MONTHNAME(order_date) AS month,
    ROUND(SUM(sales),2) AS sales
FROM retail
GROUP BY YEAR(order_date),
     MONTH(order_date),
     MONTHNAME(order_date)
ORDER BY year, MONTH(order_date);
```

Insight:

Sales are highest during the last few months of the year, especially in September, November, and December

Business Meaning:

Possible reasons include

- festival seasons
- holiday shopping
- year-end offers and discounts

5. Which months have peak sales and peak profits?

Insight:
Sales are highest during September, November, and December.

Business Meaning:
Customers tend to spend more during festive and holiday seasons.

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6. Which categories are most affected by heavy discounts?

SQL Query

```
select sub_category,
round(sum(profit),2) as loss
from retail
where discount>0.5
group by sub_category
order by loss asc
```

Insight:
Binders experience the highest losses when heavy discounts are applied.

Business Meaning:

- Large discounts reduce profit significantly.
- Selling price may become lower than business cost.

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7. Are higher quantities sold actually leading to higher profits?

Insight:
Higher quantities sold do not always lead to higher profits.

Business Meaning:
Some expensive products generate higher profits even with fewer sales.

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8. Which products should the business discontinue due to repeated losses?

SQL Query

```
select sub_category,
count(*) as loss_count,
round(sum(profit),2) as loss
from retail
where profit<0
group by sub_category
order by loss_count desc
```

Insight:

Binders and Tables generate repeated losses despite having strong sales.

Business Meaning:

- The business sells many binders.
- But discounts or low pricing reduce profitability.

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9. Which states contribute the highest revenue and profit?

SQL Query

```
select state,
round(sum(sales),2) as revenue
from retail
group by state
order by revenue desc
limit 5

select state,
round(sum(profit),2) as profits
from retail
group by state
order by profits desc
limit 5
```

Insight:

California and New York generate the highest revenue and profits.

Business Meaning:

Possible reasons include

- larger population
- higher customer demand
- stronger business activity

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10. What is the relationship between discount percentage and profitability?

Insight:

Higher discounts reduce profitability and increase business losses.

Business Meaning:

Very large discounts reduce the money earned from each product sale.

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7. Power BI Dashboard

Interactive dashboards were created to visualize customer behavior, regional performance, profitability, and sales trends.

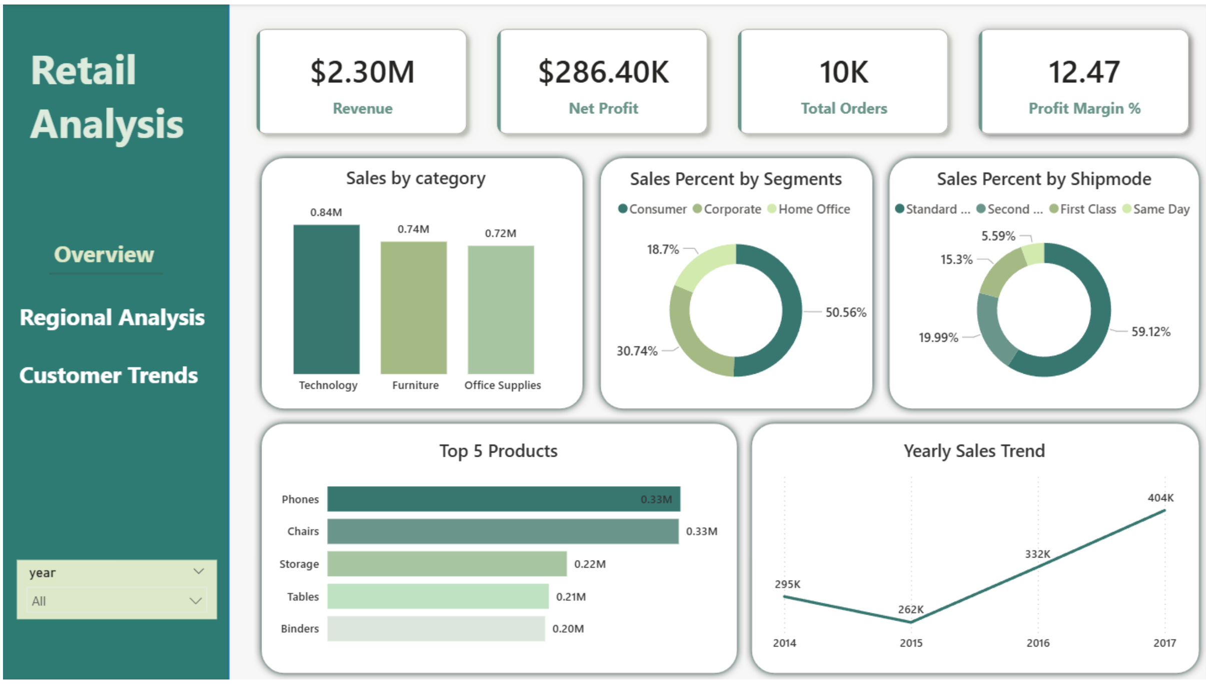

Fig 1. Business Overview

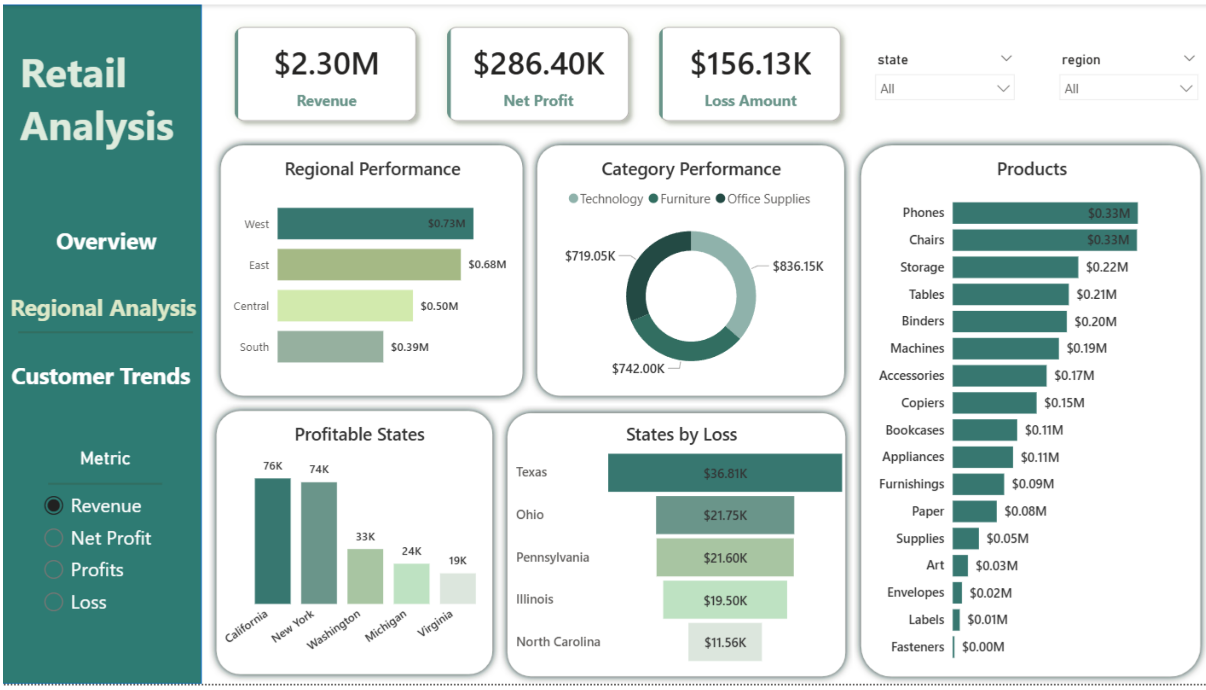

Fig 2. Regional & Profit Analysis

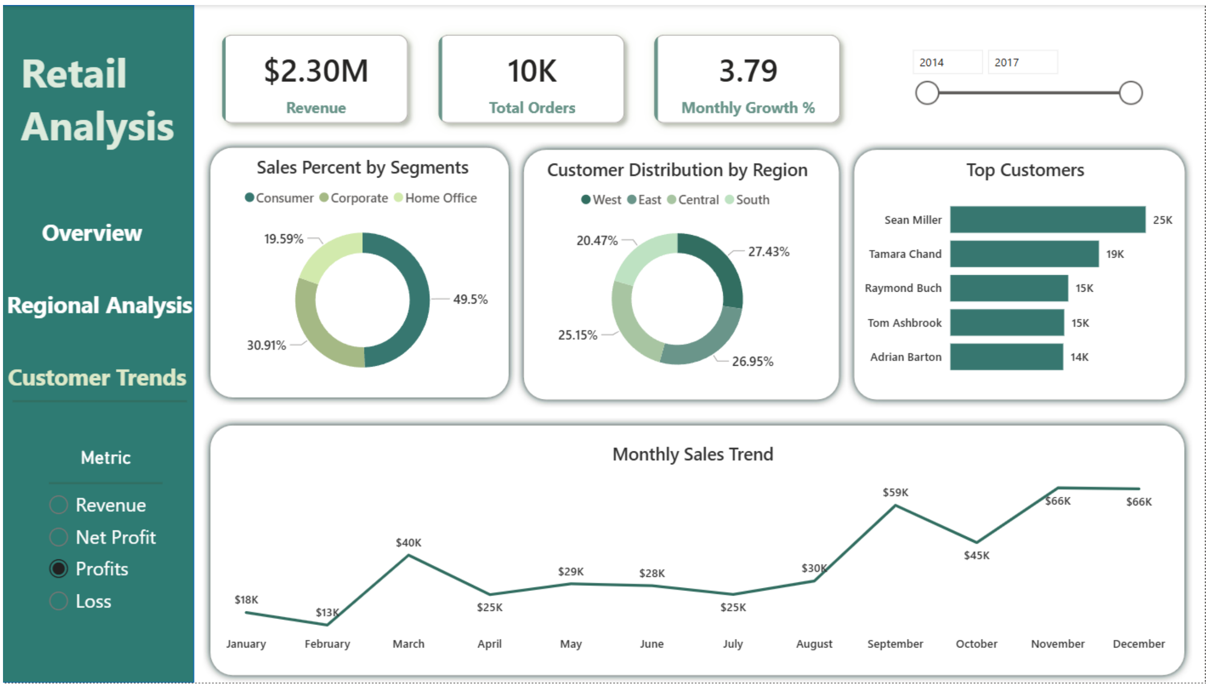

Fig 3. Customer & Sales Trend Analysis

8. Business Recommendations

- Reduce heavy discounts on loss-making products

- Focus marketing efforts on Consumer customers
- Improve pricing strategy for low-profit products
- Increase stock during peak sales months
- Optimize shipping costs for faster delivery modes

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9. Final Conclusion

This project successfully analyzed customer behavior, product performance, profitability, and sales trends using Python, SQL, and Power BI. The analysis identified high-performing customer segments, loss-making products, discount-related profitability issues, and seasonal sales patterns.

The findings can help businesses:

- Improve profitability
- Reduce losses
- Optimize discount strategies
- Focus on high-performing products and regions
- Make data-driven business decisions

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10. Tools & Technologies Used

- Python
- Pandas
- MySQL
- Power BI
- Pycharm

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